

**Department of Computer Science and Engineering**  
**Academic Year 2021– 2022 (Even Semester)**

**Degree, Semester & Branch: B.E, IV, CSE**

**Course Code & Title: CS8493 & Operating Systems**

**Name of the Faculty member (s): Dr.I.Gethzi Ahila Poornima**

**Innovative Practice Description**

- **Unit / Topic: Unit III / Page Replacement Algorithm-FIFO & Optimal**
- **Course Outcome: CO3**
- **Topic Learning Outcome: TLO2**
- **Activity Chosen: Flipped Class room**
- **Justification:**

The term "flipped classroom" refers to a method of teaching in which the traditional method of imparting knowledge is replaced by an individual self-learning process, and the traditional group activity is changed into a dynamic, interactive learning environment in which the teacher merely directs the students as they apply their creative thinking to offer a successful solution to the course material. Compared to conventional direct content transmission, it offers greater advantages.

One of the concepts that requires further explanation so that the students can comprehend it and work through the challenges associated with it is FIFO and Optimal. Group conversations helped the students comprehend the idea more fully.

- **Time Allotted for the Activity: 40 Minutes**
- **Details of the Implementation:**

The topic is selected and made public in the class four days in advance based on the justification provided. The NPTEL video and my own video lecture (<https://www.youtube.com/watch?v=MpEpDBEF4tM>) have already been shared in the previous class and the students are asked to watch it and be prepared for the group discussion. They are also asked to make the appropriate notes. Small groups (size as 3) are created and given topics to discuss with the other members of the peer team on the activity day. At the end of the session, one group was called to present on board. Figure 1 displays a few representative pictures from the event that were captured.

- **CO – PO / PSO mapping:**

| CO  | PO2 | PO4 | PSO1 |
|-----|-----|-----|------|
| CO3 | 2   | 3   | 2    |

(1 – Low      2 – Moderate      3 – High)

- **PO / PSO mapped:**

| Innovative practice                  | PO2                                                                                                   | PO4                                                                                             | PSO1                                                                                                                                  |
|--------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
|                                      | 2                                                                                                     | 3                                                                                               | 2                                                                                                                                     |
| <b>Justification for correlation</b> | Students could identify appropriate page replacement algorithm for an Operating Systems and apply it. | Students will be able to design a suitable page replacement algorithms for an Operating system. | Students will be able to analyze the performance of the system while applying page replacement algorithms and develop new algorithms. |

- **Images / Screenshot of the practice:**



**Figure 1:** Screenshots during the activity

- **Reflective Critique:**

- ❖ *Feedback of practice from students and other stakeholders:*

- The reaction from the students regarding this activity was excellent. They believed that by engaging in this activity, they would be able to explore this idea more thoroughly.

❖ ***Benefit of the practice:***

- They learned a few things separately that led to self-learning. Learning at home naturally becomes student-centric rather than teacher-centric.
- They participated actively in this meeting rather than having a one-way conversation.

❖ ***Challenges faced in implementation:***

- A few students had trouble understanding the notion, but they overcame it by talking to their teammates.

**Signature of Faculty Member**

**HOD**